

- 4 The transport systems of plants and mammals both function to transport substances to and from cells.

- (a) Table 4.1 contains descriptions concerning the mammalian circulatory system and the structure of the heart.

Complete Table 4.1 by writing down the term that matches each description.

Table 4.1

description	term
a transport system where blood is enclosed in blood vessels and passes through the heart twice in one complete circuit of the body	
blood vessel, with a lumen of approximately $7\mu\text{m}$ in diameter, that supplies substances to cells	
the blood vessel that carries oxygenated blood from the lungs to the heart	
the chamber of the heart that receives deoxygenated blood from the body	
the structure dividing the heart into the left and right sides	

[5]

- (b) Water is the main component of the fluids associated with transport in mammals and plants.

- (i) One property of water is its ability to act as a solvent.

Suggest why this property is important in the mammalian transport system.

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- (ii) The transport of water in plants depends on the ability of water molecules to form hydrogen bonds.

Explain how hydrogen bonding is involved with the movement of water in the xylem.

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[Total: 10]